

Orientation and report writing

Fall 2026 · Prof. Ehsan Roohi
September 9, 2026

**MEASURE
MODEL
COMPARE**

Experimental evidence and engineering judgment

01

OBSERVE

See the physical behavior
worth explaining

02

MEASURE

Acquire data with a
deliberate measurement
system

03

COMPARE

Test data against theory
and models

04

COMMUNICATE

Write a report another
engineer can trust

The goal is not merely to collect numbers. The goal is to make a claim that the data support.

Today's 50-minute session

- 01** 0–8 min: course organization and workflow

- 02** 8–23 min: experiments, apparatus, and analysis

- 03** 23–42 min: report writing and comparison

- 04** 42–48 min: policies and report exercise

- 05** 48–50 min: first dates and preparation

People, places, and weekly rhythm

COURSE TEAM

Prof. Ehsan Roohi
Yuyuan Zhang
Nazila Emamdoost

WEEKLY RHYTHM

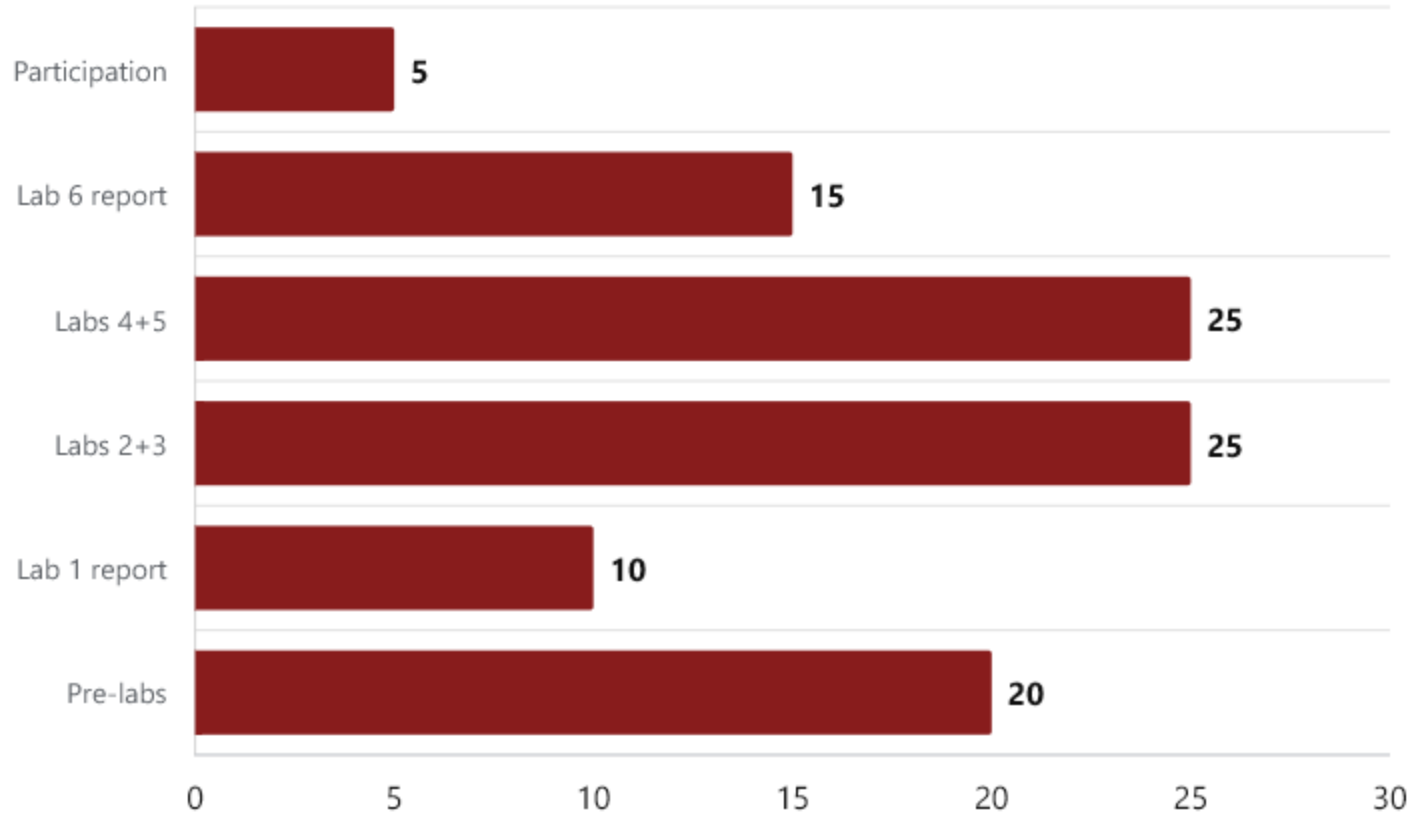
MON Theory • 12:20–1:10 PM
Hasbrouck Laboratory 134

TUE–FRI Laboratory sections
Gunness Laboratory 5

FRI Office hours • 12:20–1:10 PM
Gunness Laboratory 1

Check Canvas for your assigned section, GTA, and any section-specific updates.

Course grading

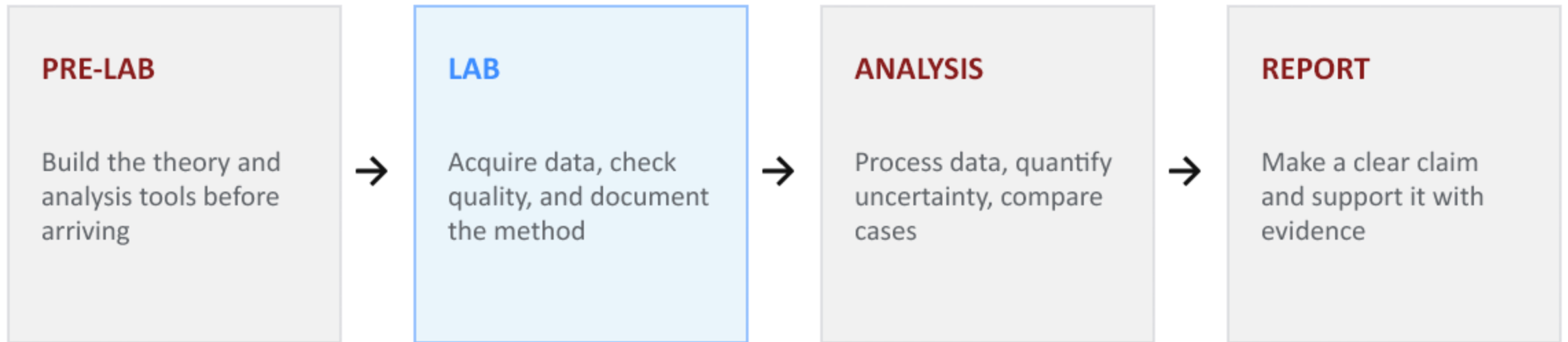


100%

**Preparation + reports +
participation**

The combined reports reward
synthesis across related
experiments.

The laboratory workflow



Pre-labs are due before your own laboratory section begins.

Six experiments and four reports

Lab	Experiment	Primary focus	Report
1	Dynamic data sampling	Sampling, aliasing, spectra	Lab 1
2	Single pendulum	Dynamic response, model–data	Labs 2+3
3	Double pendulum	Coupled and nonlinear motion	Labs 2+3
4	1-DOF free response	Natural frequency, damping	Labs 4+5
5	1-DOF forced response	Resonance, frequency response	Labs 4+5
6	Radiometer	Movement measurement	Lab 6

Lab 1: Sampling and frequency analysis

Apparatus

Moku:Go data logger, microphone, speaker, and sound sources such as a tuning fork.

Data collection

Record at least one second at several rates relative to the source frequency, then replay and save each data file.

Analysis

Compare time histories and FFT spectra. Identify the fundamental frequency, aliasing, and the effect of record length.

Report focus

Explain why sampling rate controls representation while record duration controls frequency resolution.

$$f_s > 2 f_{\max} \quad \Delta f = f_s / N = 1 / T_{\text{record}}$$

The procedure includes rates such as 100f, 10f, 3f, 1.4f, and 1f. Record the actual settings for every run.

Lab 2: Single pendulum

Apparatus

Single pendulum, high-speed camera at the assigned frame rate, scale reference, and the MATLAB tracking app.

Data collection

Release from rest at small, medium, and large initial angles. Record the angle, geometry, masses, and video settings.

Analysis

Convert tracked coordinates to angle and angular velocity. Compare time histories, spectra, period, and numerical prediction.

Report focus

Explain when the small-angle model works and how inertia, damping, calibration, and release angle affect agreement.

$$I_p \ddot{\theta} + m g d \sin \theta = 0 \quad T_{\text{small}} = 2\pi \sqrt{I_p / m g d}$$

The procedure uses releases below 10°, from 10° to 90°, and from 90° to 180°. Match simulation and experiment initial conditions.

Lab 3: Double pendulum

Apparatus

Two-link pendulum, high-speed camera, scale reference, and image tracking for both moving masses.

Data collection

Record a small-angle release and at least three large-angle cases. Save both initial angles and release conditions.

Analysis

Compute both angles and angular velocities, then compare time histories and spectra with the coupled numerical model.

Report focus

Discuss coupling, sensitivity to initial conditions, tracking error, and where measured and simulated trajectories begin to diverge.

A complicated or diverging trajectory alone does not establish chaos. First verify initial conditions, time alignment, and tracking.

Lab 4: Free response of a 1-DOF system

Apparatus

Vernier cart and position sensor, force sensor, springs, added masses, and eddy-current dampers.

Data collection

Measure mass and spring stiffness. Displace the cart, release it, and repeat with changed mass, stiffness, and damping.

Analysis

Use displacement histories, peak decay, and spectra to estimate damped frequency, natural frequency, and damping ratio.

Report focus

Relate each controlled change to the model and document effective stiffness, initial displacement, sampling rate, and peak selection.

$$m \ddot{x} + c \dot{x} + k x = 0 \quad \omega_n = \sqrt{k / m} \quad \zeta = c / (2\sqrt{k m})$$

The procedure uses a base case plus variations in mass, springs, and dampers. Verify each exported CSV before leaving.

Lab 5: Forced response of a 1-DOF system

Apparatus

Vernier cart, springs, motor-driven input, magnetic dampers, tachometer, and Graphical Analysis software.

Data collection

Record three impulse cases, then harmonic steady-state responses below and above resonance while avoiding prolonged resonance.

Analysis

Separate transient and steady motion. Compute output-to-input magnitude and compare the experimental Bode plot with the model.

Report focus

Use measured input amplitude and frequency for every point. Connect Lab 4 parameter estimates to the forced-response prediction.

$$m \ddot{x} + c \dot{x} + k x = F(t)$$

Let the response settle before measuring amplitude. Convert tachometer RPM to rad/s and document any configuration changes.

Lab 6: Radiometer rotation

Apparatus

Crookes radiometer, LED light, lux meter, fixed camera, matte black background, ruler, and optional IR thermometer.

Data collection

Record a clear 60–90 s video at a close safe light distance. Keep geometry fixed and note lux and thermal history.

Analysis

Track the vane within an annulus, unwrap cumulative angle, fit its slope for average RPM, and inspect interval RPM for slowdown.

Report focus

Connect rotation to rarefied-gas thermal transport. Explain why lux is an imperfect heat-input measure and why warming matters.

$$\theta(t) \approx \theta_0 + \omega_{\text{avg}} t \quad \text{RPM} = (60 / 2\pi) \omega_{\text{avg}}$$

A practical start is about 15 cm from the light with the background 20–30 cm behind the bulb. Follow the released safety limits.

The structure of a laboratory report

REQUIRED SECTIONS

Abstract

1. Introduction

2. Method

3. Results

4. Comparison

5. Discussion

6. Conclusions

References • Appendices

WHAT THE READER SHOULD FIND

- A clear purpose and relevant theory
- A reproducible method
- Readable figures with units and labels
- Direct comparison with theory or a model
- A discussion of discrepancies and limitations
- Conclusions supported by the presented data

Meaningful results matter more than a polished page with weak analysis.

Report length and section purpose

Section	Template allocation	Main purpose
Abstract	Maximum 10 lines	Purpose, method, key finding, implication
Introduction	1 page	Theory, experiment, and testable objective
Method	1 page	Enough detail to reproduce the measurement
Results	3 pages including figures	Observed behavior and processed measurements
Comparison	1 page	Prediction and experiment on the same basis
Discussion	1 page	Discrepancies, uncertainty, and limitations
Conclusions	1 page	Quantitative answer and the main lesson

References and optional appendices have no stated page limit. Follow any assignment-specific limit on Canvas.

Introduction and reproducible method

Introduction

State a testable objective. Give the governing equation, define symbols, and explain assumptions.

Method

Describe the actual apparatus, calibration, sampling settings, initial conditions, and repeated runs.

Analysis record

Explain processing choices, fitting intervals, and parameter sources so another person can repeat the analysis.

Example objective: test how release angle changes agreement with the small-angle pendulum model.

A concise, quantitative abstract

Purpose and method

We tested a small-angle pendulum prediction using angle histories from video.

Key result

The measured frequency was 1.47 Hz, compared with a predicted 1.50 Hz, a 2.0% difference.

Meaning and limit

The agreement supports the model for this case. The discrepancy requires an uncertainty assessment.

Illustrative wording and invented teaching values. An actual abstract must use your own results. Maximum 10 lines.

Results: the measured behavior

Observation

Describe what the measurements show, with quantities and units.

Evidence

Use figures or tables that answer the experimental question. Identify conditions and repeated runs.

Traceability

Distinguish measured data, processed measurements, and simulation predictions.

Example observation: successive positive displacement peaks decrease over time.

Readable, exported figures

DO

- Export high-resolution PNG, EPS, or PDF
- Label axes and include units
- Use legends that identify each data set
- Group related cases when comparison improves the analysis

AVOID

- Screenshots of MATLAB or Python figure windows
- Tiny labels and unreadable legends
- Decorative plots that do not answer a question
- Separate figures when one comparison plot is clearer

A figure should still make sense when viewed without the surrounding paragraph.

A figure that supports a comparison

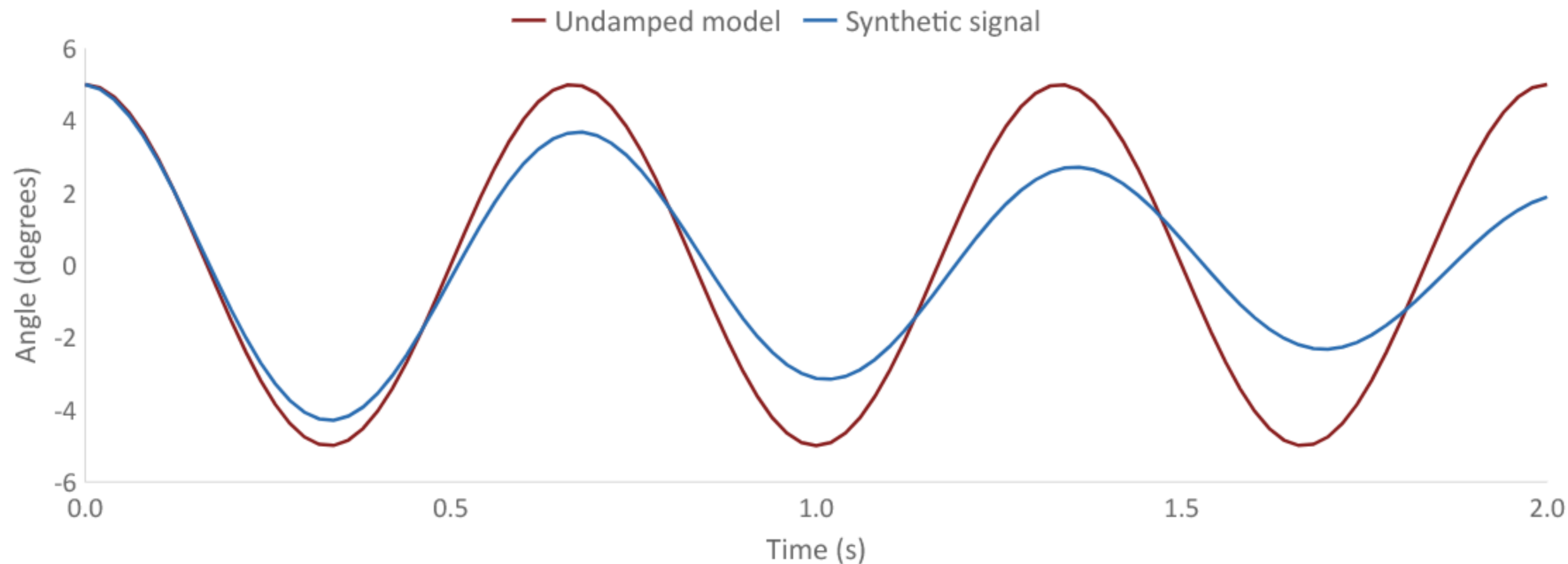


Figure 1. Illustrative angle histories: undamped model and synthetic decaying signal, 5° initial angle, 50 samples/s.

Comparison: a fair test of the model

Same basis

Use matching units, initial conditions, time origin, and system parameters. Explain the theoretical solution.

Quantitative comparison

Compare period, frequency, amplitude, phase, damping, or another quantity relevant to the objective.

Parameter honesty

State which parameters you measured, assumed, or fitted. A fit to the same data is not independent validation.

$$\text{Relative discrepancy} = | \text{measured} - \text{predicted} | / | \text{predicted} | \times 100\%$$

Use an appropriate absolute or scaled metric when the predicted value is zero or very small.

Uncertainty and acceptable agreement

Measurement uncertainty

Consider calibration, resolution, repeatability, timing, and tracking choices.

Model uncertainty

Consider uncertain parameters and omitted physics, such as friction or large-angle effects.

A justified conclusion

Report what each uncertainty means. Compare the discrepancy with a consistent uncertainty estimate.

$$T = \Delta t / n_{\text{cycles}} \quad uT \approx u\Delta t / n_{\text{cycles}}$$

Simplified timing example with exact cycle count. Longer timing intervals do not remove calibration bias.

Discussion: explanations tied to evidence

Observed difference

The measured oscillation envelope decays while the undamped prediction keeps constant amplitude.

Plausible mechanism

Friction or drag could dissipate energy. The present comparison does not identify their separate contributions.

Discriminating check

Compare decay across configurations and test whether the chosen damping model describes the envelope.

Replace “human error” with a specific mechanism, its expected effect, and a way to check it.

Conclusions, references, and appendices

Conclusions

Answer the objective with the key numerical result, supported limitation, and a useful improvement.

References

Cite sources where used. Include the lecture notes at minimum and list every cited source.

Appendices

Add supporting calculations or extra data only when needed. Keep essential evidence in the main text.

Combined reports connect related experiments

Labs 2 and 3

Compare one angle with two coupled angles. Discuss model assumptions, initial conditions, and the limits of prediction.

Labs 4 and 5

Relate free-response frequency and damping to the forced response. Account for configuration changes.

Organization

Use one abstract and conclusion, with experiment-specific subsections and an explicit synthesis.

Suggested organization within the required template. Do not assume the page allowance doubles.

The report rubric: ten equal criteria

Each criterion is 10% of the report grade	What to check
Complete sections / overall aesthetic	Required organization and consistent formatting
Clear figures / abstract	Readable evidence and a concise, quantitative summary
Data analysis / introducing the theory	Traceable processing and defined governing equations
Introducing the experiment / comparison with theory	Physical context and a fair model-data comparison
Discussion / meaningful results	Evidence-based interpretation and an answer to the objective

This table groups the ten original rubric criteria into pairs. Each individual criterion remains worth 10%.

Submission and integrity policies

DEADLINES

No late pre-labs.
Reports lose 25% for each day or partial day late.

MAKE-UPS

Require instructor/GTA availability, space, and the correct equipment setup.

INTEGRITY

Discuss approaches, but do not copy solutions or share MATLAB, Python, or Simulink files.

VERIFICATION

Any pre-lab or report may include a brief oral check of your understanding.

Notify the instructor during the first two weeks about approved conflicts or accommodation needs.

Three-minute report exercise

Draft sentence

“The frequency was 1.47. Theory was 1.50. The error was small because of human error.”

Work with a neighbor

Add units, calculate the discrepancy, and separate the observation from its interpretation.

Then decide

What information is missing before you can judge agreement? Suggest one check that would help.

Invented teaching values. No submission is required for this in-class discussion.

Exercise debrief

Results and comparison

The measured frequency was 1.47 Hz and the predicted frequency was 1.50 Hz. Their difference was 0.03 Hz, or 2.0% of the prediction.

Discussion

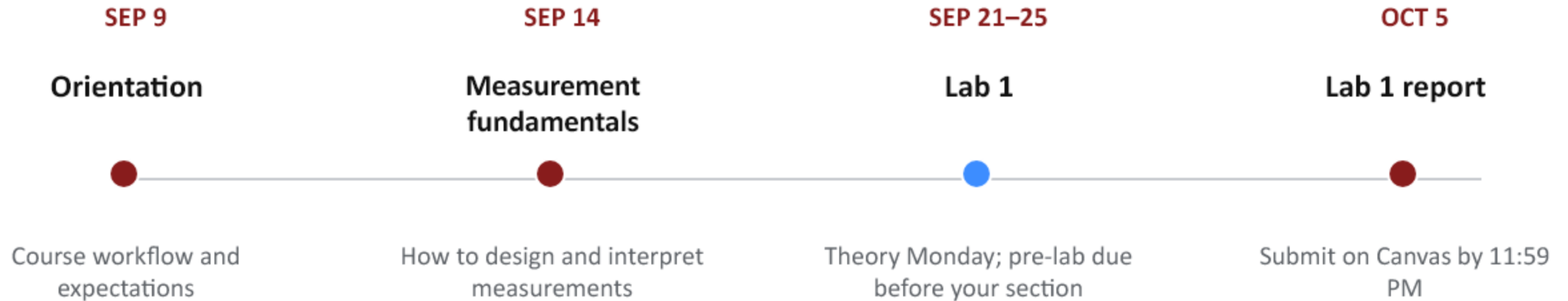
This discrepancy alone does not establish acceptable agreement. Timing and model-parameter uncertainties are needed.

Useful next check

Verify the time calibration and repeat the frequency estimate using a stated interval and method.

Other specific, evidence-based checks can also be valid.

The first dates



There is no laboratory in Weeks 1–2.

Preparation for Lab 1

BEFORE WEEK 3

- Confirm your laboratory section and GTA on Canvas
- Install or update MATLAB and Simulink
- Open the Lab 1 instructions and Pre-lab 1
- Know where and when your section meets
- Bring a way to save your experimental data

ONE QUESTION

**What would make
you trust an
experimental result?**

Questions & discussion